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My design focused on usability, feasibility, and desirability.

I wanted the screwdriver to be easy to hold, easy to understand, and realistic to build as a prototype. The assignment asked for a handheld electric screwdriver with a variable speed control, a non-digital direction control, a way to choose between 10 screwdriver heads, and a digital screen that shows how many screws were screwed in and unscrewed. It also had to work for both right- and left-handed users and include a rechargeable battery.

To meet these requirements, I made three design ideas in my sketches.

In the first sketch, I put the screen on the top so the user can quickly see the screw count. I added a lever for speed control and placed the battery near the bottom for better balance. I also showed a space in the body to store the 10 screwhead choices. For direction, I used a twist left or right action to screw or unscrew.

In the second sketch, I tried a more horizontal form. This version also stores the 10 screwhead choices in the body. I placed the battery in a way that could help with balance and make battery replacement faster. I also explored using the trigger pressure to change speed.

In the third sketch, I explored a design with a transparent cover so users can see the stored screwheads. I kept the screen on the side, the battery in the handle, and a clear speed selection control.

I designed for more than one goal. For usability, I tried to place controls where they are easy to reach and understand. For feasibility, I thought about where the battery and screwdriver heads could fit inside the product. For desirability, I wanted the tool to look modern, simple, and easy to use, not too bulky or confusing.

Prototype

I made low-fidelity prototypes based on my sketches. The goal of the prototype was not to make a fully working screwdriver, but to test the shape, control placement, and overall interaction.

My process started with brainstorming and sketching different ideas. After that, I compared the three concepts and looked at which one best met the assignment requirements. Then I chose the strongest direction and developed it further into a physical prototype.

In my prototype, I focused on:

where the user would hold the screwdriver where the speed control should go where the direction control should go where the digital screen should be placed how the 10 screwheads could be stored in the product where the battery should be placed for better balance

The full process was:

read the design requirements create several quick concept sketches compare the concepts choose one main direction make a low-fidelity physical prototype test the prototype with a user reflect on what worked and what should change

Analysis

What worked well:

The overall idea of storing the 10 screwheads inside the body was clear and useful. The screen helped communicate screw count clearly. The battery placement idea made sense for balance. The general form looked compact and understandable as a handheld tool.

What needed improvement:

The twist left or right control for screw/unscrew may not be the easiest or most comfortable method during real use. The speed control needs to be clearer. It should be obvious whether the user should press harder, move a lever, or switch between set levels. The way the 10 screwdriver heads switch in and out needs more explanation and maybe a clearer mechanism. The screen placement may need adjustment so it is easier to see while holding the product.

From the testing, I concluded that the design idea is effective as a starting concept because it addresses the main assignment requirements and shows a clear relationship between the physical controls and the digital display. However, the prototype also showed that the control system needs to be simplified and made more intuitive. The next step would be to improve the direction control, make the speed control more obvious, and better explain how the stored screwdriver heads are selected and used.